IN6299/IS6799/KM6399 CRITICAL INQUIRY

Research Proposal:

Analysis of a Large Dataset Using Polars Open-Source Library

Group ID: VH-03-02

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Content

1. Introduction

As the fast development of e-commerce platforms, a large amount of interaction data was generated continuously, such as page views, add-to-cart operations and purchase records. The "RecSys 2020 Preprocessed eCommerce Behavior Dataset" is a dataset from a mult-category online store, which captures these kinds of events from October 2019 to April 2020. This dataset covers session-level interactions, product information and time attributes. It also adopted Parquet format to support high efficiency storage, retrieval and large scale aggregation. It provides a comprehensive foundation for analyzing behavior patterns and building recommendation-oriented insights due to its integration of session, users and products.

It is a huge challenge to efficiently process such a large datasets. Traditional Python data analysis frameworks like Pandas often face problems such as high memory usage and slow execution speed when processing large scale aggregation and time-based analysis. Polars is an open-source DataFrame library based on Rust, which is specially designed for high performance and large scale data processing. It supports parallel computing and lazy evaluation. This study is aimed at evaluating the real efficiency of Polars in business e-commerce data analysis. By using Polars, this study will not only demonstrate a faster and scalable data processing ability, but also extract business insights, such as user interaction patterns, popular products and product co-occurrence relationships. These insights will provide decision making strategies for e-commerce platforms.

This proposal will focus on the performance of Polars under the environment of large scale e-commerce behavior. The study will include systematic data preprocessing, baseline exploratory data analysis, user behavior pattern analysis and recommendation-oriented analysis. We are also going to compare the performance metrics including execution time, memory usage as well as scalability, which will demonstrate the advantages of Polars under large datasets. Ultimately, the research aims to show a reproducible analytical workflow which can integrate high performance data processing with operable business insights. The workflow will demonstrate how Polars support efficient computation and meaningful behavior analysis on large-scale datasets.

2. Problem Statement & Research Motivation

For the usability and maturity of its environment, Pandas is a commonly used tool when conducting data analysis tasks. However, because it is not built specifically for large datasets, problems like slow processing speed and high RAM usage often occur. These cases are especially prevalent when conducting projects that involve grouping, aggregation, and time-series analysis.

This research will address this problem by adopting the data frame library Polars for Python. This tool is built for large-scale data analytical scenarios by supporting parallel computing. By comparing the efficiency of Pandas and Polars in tasks such as data processing, cleaning, and grouping, etc., the research could demonstrate Polars' obvious advantage in dealing with massive datasets. In addition, conduct user behaviour analysis and export business insights from the chosen dataset, to further illustrate how different tool choices could affect analytical practices.

3. Aims & Objectives

This research aims to compare Pandas and Polars' performance when conducting analytical tasks for large-scale E-Commerce datasets. *The RecSys 2020 Preprocessed eCommerce Behaviour Data From Multi-Category Store Dataset* was chosen as the target for this study because it contains millions of user interaction data, which is suitable for representing a typical massive dataset (Schettler, 2021).

The objectives include:

1. **Building a standardized analytical framework** to support systematic studies on E-commerce user data.
2. **Data pre-processing using Pandas and Polars**, such as missing and outlier values, filtering, and splitting the data into subsets.
3. **Conducting fundamental exploratory data analysis** to support later data visualization and summarization.
4. **Performing User behavior pattern analysis** based on user action patterns.
5. **Using association rule and itemset mining** techniques to identify shopping trends and support promotion mechanisms.
6. **Assessing Polars' advantage** in large-scale data analysis by demonstrating its processing speed and capability.

4. Methodology

4.1 Overall Analytical Framework

The methodology of this research is designed around two core objectives:

1. Systematically compare the performance of Pandas and Polars in processing large-scale e-commerce behavior data.
2. Use polar to effectively analyze user behavior to extract meaningful business insights.

4.2 Data Preprocessing

Before analysis, the train.parquet file will be preprocessed to ensure data quality and consistency. The main steps include:

- Handling missing values and incorrect records (e.g., missing brand information or invalid price data)

- Transformation and standardization of time-related fields (e.g., timestamp parsing and validation of derived temporal features)
- Filtering data and constructing subsets according to analysis needs.
- Dividing the dataset into subsets of different sizes to facilitate the comparison of Pandas and Polar's performance under different data scales.

All preprocessing processes will be implemented using both Pandas and Polars to build a consistent and controlled foundation for subsequent performance evaluation.

4.3 Baseline Exploratory Data Analysis (EDA)

To fully understand the structure and distribution of the dataset, a baseline EDA will be performed. The tasks in stage include:

- Statistical analysis of user behavior types (e.g., distribution of different event types).
- Analysis of session length distribution.
- Analysis of interaction intensity of different product categories
- Temporal trend analysis of user behavior.

These analyses provide a macro-level understanding of the data characteristics and also serve as baseline cases for demonstrating the efficiency of PolarDB in large-scale aggregation and statistical operations.

4.4 User Behavior Pattern Analysis

Besides to the baseline EDA, this study will further analyze more structured user behavior patterns, such as:

- **Popular Products and Categories Analysis:** Analyze the frequency of add-to-cart and purchase events across products and categories in the data.
- **User activity analysis:** Check the frequency of interaction between users, and compare the behavioral differences between new and old users.
- **Information visualization:** Identifying potential patterns and trends thru data visualization.

This stage involves complicated grouping, sorting, and aggregation operations based on session and time dimensions, which can well highlight the advantages of polars in complex behavior analysis tasks

4.5 Recommendation-Oriented Analysis

Given the recommendation system context of the dataset, this study will conduct a variety of recommendation-oriented analysis without building complex machine learning models. Specifically:

- **Association rule mining:** Analyze the co-occurrence relationship of products in users or sessions to identify goods that are often viewed or purchased together, and simulate simple recommendation logic. Apriori, FP-Growth or basic frequent itemset mining techniques can be used for this purpose.

4.6 Pandas vs Polars Performance Comparison

For the purpose of to systematically evaluate the performance differences between the two data processing frameworks, the same analysis tasks will be implemented using Pandas and Polars, including:

- Data loading and preprocessing
- Session-based and time-based aggregation statistics
- Behavior distribution and trend analysis

Performance evaluation will consider, but is not limited to the following metrics:

- Execution time
- Memory usage
- Scalability under large data volumes
- Implementation complexity and code readability

5. Expected Outcomes

- Polars demonstrates high efficiency and scalability in the aspect of large scale data processing. These advantages allow data analysts and engineers to shorten processing time and reduce computational costs.
- From EDA and user pattern analysis, products manager and user experience teams could optimize user journeys and improve platform experience.
- The analysis of product and category may help making decisions in inventory planning and promotion strategy.
- The recommendation designers and marketing teams can discover the frequent co-occurring products from recommendation-oriented analysis.
- By getting full-scale user behavior data, the business analysts will be able to make timely and data driven decisions.

6. Project Schedule

Key Tasks	Week1 - Week4		Week 5/6	Week 7/8	Week 9/10	Week 11/12	Week 13/14	Week 15
Phase 1								
Proposal preparation								
Proposal Submission(Deadline: 4 feburary)								
Phase 2								
Data preprocessing (Pandas & Polars)								
Pandas vs Polars performance benchmarking								
Baseline EDA and user behavior analysis								
Recommendation-oriented analysis								
Phase 3								
Result interpretation and report writing								
Final report submission (Deadline: 28 April)								

7. References

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